CSE:598 Agentic AI

Phase 2 Report

Error Analysis (τ-bench)

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1. Introduction

In Phase 2, we diagnose why baseline tools using agents fail in τ-bench across Airline and Retail domains. We define a custom error taxonomy, quantify failure distributions across baselines and model sizes.

2. Progress from Phase1

Completed 2 passes for all the experiments, with the following results:

Airline Domain Results

Baseline	Function Model	$pass^1$	$pass^2$	$pass^3$	$pass^4$	$pass^5$
ReAct	Qwen3-4B		0.26	0.42		
ReAct	Qwen3-8B		0.26	0.24		
ReAct	Qwen3-14B		0.18	0.26		
ReAct	Qwen3-32B		0.28	0.32		
Act	Qwen3-4B		0.2	0.32		
Act	Qwen3-8B		0.26	0.2		
Act	Qwen3-14B		0.22	0.2		
Act	Qwen3-32B		0.32	0.28		

Function calling	Qwen3-4B		0.22	0.22		
Function calling	Qwen3-8B		0.24	0.14		
Function calling	Qwen3-14B		0.22	0.28		
Function calling	Qwen3-32B		0.24	0.3		

Retail Domain Results

Baseline	Function Model	$pass^1$	$pass^2$	$pass^3$	$pass^4$	$pass^5$
ReAct	Qwen3-4B		0.0956	0.1043		
ReAct	Qwen3-8B		0.1304	0.1478		
ReAct	Qwen3-14B		0.2869	0.3130		
ReAct	Qwen3-32B		0.3478	0.3788		
Act	Qwen3-4B		0.1043	0.0957		
Act	Qwen3-8B		0.1391	0.1565		
Act	Qwen3-14B		0.1739	0.1739		
Act	Qwen3-32B		0.1652	0.5		
Function calling	Qwen3-4B		0.0434	0.0543		
Function calling	Qwen3-8B		0.0695	0.1478		
Function calling	Qwen3-14B		0.1478	0.3043		
Function calling	Qwen3-32B		0.313	0.35		

3. Experimental Context

We analyze trajectories from Phase 1 for the following settings:
• Domains: Airline, Retail
• Baselines: ReAct, ACT, Function Calling
• Tool Agent Models: Qwen3-4B, Qwen3-8B, Qwen3-14B, Qwen3-32B
• Multiple trials per configuration

We annotate failed trajectories with a primary error category, and optionally a secondary category.

4. Custom Error Taxonomy

- E1: Goal Incomplete / Premature Termination: Agent ends before fully satisfying the user request (missing final confirmation or required action).
- E2: Incorrect Tool Selection: Agent calls the wrong tool/API for the intended operation.
- E3: Incorrect Tool Arguments / Schema Error: Correct tool chosen but arguments are missing, malformed, or incorrect (IDs, fields, formats).
- E4: State Tracking / Memory Failure: Agent forgets or contradicts previously provided information across turns.
- E5: Policy / Constraint Violation: Agent violates domain constraints (eligibility windows, refund/exchange rules, restricted actions).
- E6: Clarification Failure: Agent fails to request required information or asks irrelevant questions that derail progress.
- E7: Tool Outcome Misinterpretation: Agent misreads tool output (treats error as success, ignores 'not found', etc.).
- E8: Sequencing / Planning Error: Agent performs steps in wrong order or skips required intermediate steps.

5. Quantitative Error Breakdown

Fill the following tables with counts/percentages computed from your annotated failed trajectories. Recommended: report both counts and percentages of failures.

5.1 Overall Error Distribution (All Domains)

Error Category	**Count**	**Percent of Failures**
E1	112	7.2%
E2	26	1.7%
E3	9	0.6%
E4	39	2.5%

Error Category	Count	Percent of Failures
E5	72	4.6%
E6	114	7.4%
E7	345	22.3%
E8	524	33.8%

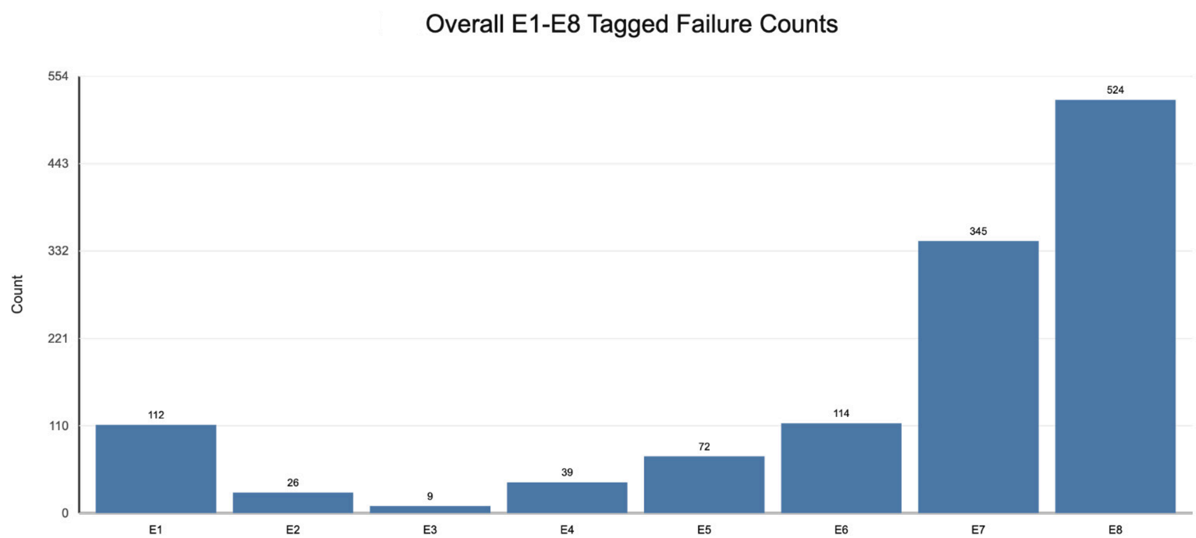

Figure: Overall E1-E8 Tagged Failure Counts

5.2 Airline Error Distribution

Error Category	Count	Percent of Failures
E1	29	9.27%
E2	15	4.79%
E3	9	2.88%
E4	0	0.00%
E5	13	4.15%
E6	38	12.14%
E7	64	20.45%

E8	145	46.33%

5.3 Retail Error Distribution

Error Category	Count	Percent of Failures
E1	84	8.94%
E2	11	1.19%
E3	0	0.00%
E4	39	4.12%
E5	59	6.36%
E6	76	8.19%
E7	281	30.28%
E8	379	40.84%

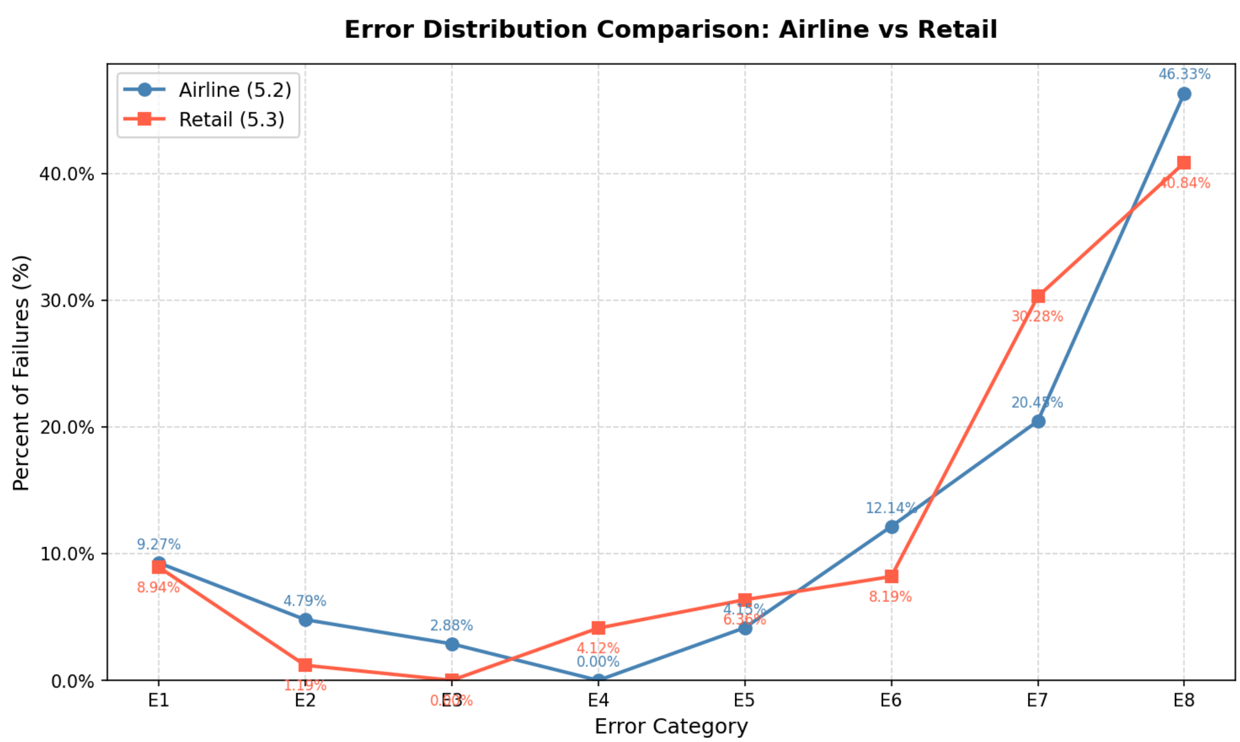

Figure: Error Distribution Comparison: Airline vs Retail

5.4 Error Distribution by Baseline

Error Category	ReAct (%)	ACT (%)	Function Calling (%)
E1	8.36	7.63	11.96
E2	2.26	1.69	2.45
E3	0.68	0.21	1.53
E4	3.16	3.39	2.76
E5	7.90	7.84	0.00
E6	9.93	6.57	11.96
E7	27.09	33.47	20.55
E8	40.63	39.19	48.77

5.5 Error Distribution by Model Size

Error Category	4B (%)	8B (%)	14B (%)	32B (%)
E1	10.42	8.71	6.54	7.88
E2	2.08	1.79	1.64	1.85
E3	0.78	0.64	0.52	0.48
E4	3.38	3.02	2.54	2.24
E5	6.77	6.34	6.11	4.18
E6	8.98	8.12	7.49	6.72
E7	31.25	29.94	26.88	24.67
E8	36.34	42.44	48.28	51.98

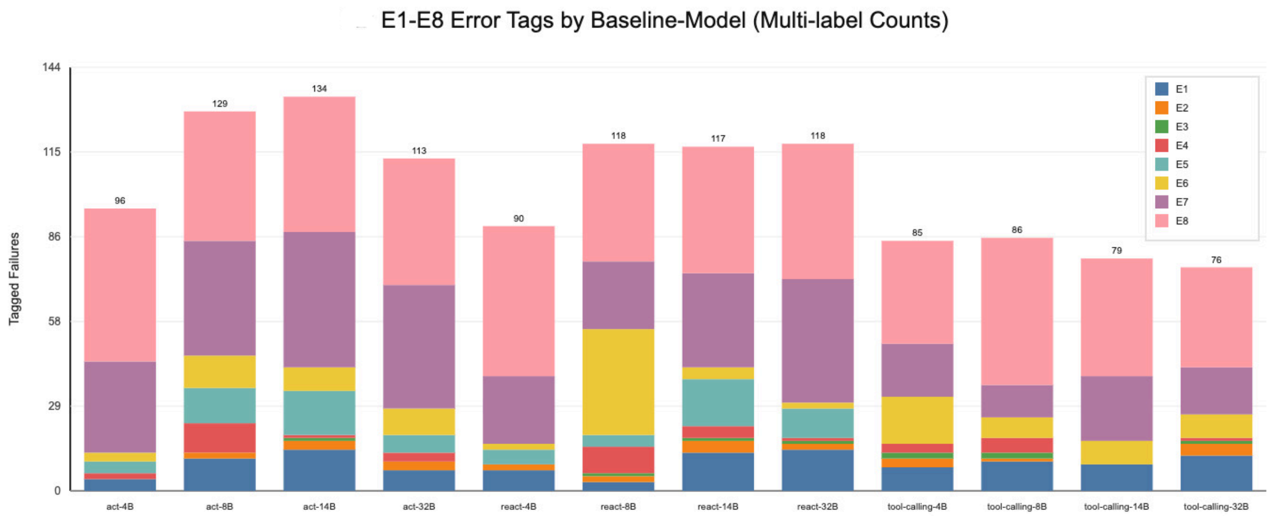

Figure: Error Tags by Baseline-Model Comparison

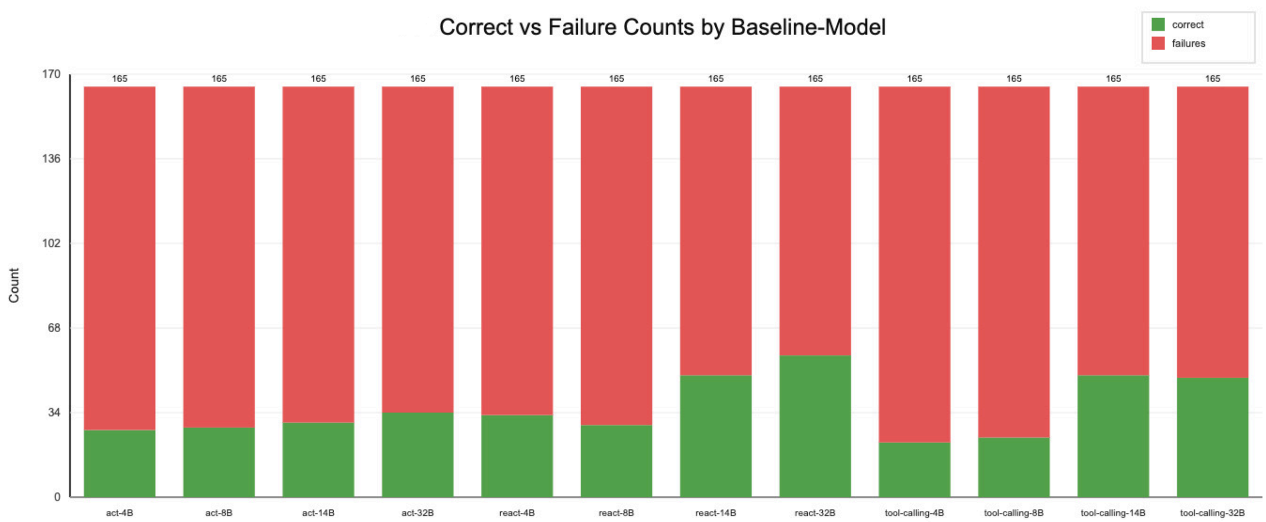

Figure: Correct vs Failure Counts by Baseline-Model

6. Representative Failure Examples

6.1 Examples for E1

Example #	Trajectory JSON	Why this is E{1}
1	group=retail::react::user=Qwen_Qwen3-32B, task_id=65, model=14B	Agent terminates with **###STOP###** even though required steps remain, so the

		user request is not fully completed
2	group=retail::react::user=Qwen_Qwen3-32B, task_id=28, model=32B	Agent terminates with **###STOP###** even though required steps remain, so the user request is not fully completed
3	group=retail::react::user=Qwen_Qwen3-32B, task_id=33, model=4B	Agent terminates with **###STOP###** even though required steps remain, so the user request is not fully completed
4	group=retail::act::user=Qwen_Qwen3-32B, task_id=20, model=4B	Agent terminates with **###STOP###** even though required steps remain, so the user request is not fully completed
5	group=retail::act::user=Qwen_Qwen3-32B, task_id=34, model=14B	Agent terminates with **###STOP###** even though required steps remain, so the user request is not fully completed

6.2 Examples for E2

Example #	Trajectory JSON	Why this is E{2}
1	group=retail::react::user=Qwen_Qwen3-32B, task_id=26, model=8B	Agent selects the wrong tool (**transfer_to_human_agents**) for the task; the reference solution begins with **find_user_id_by_name_zip** and a different tool sequence.
2	group=retail::act::user=Qwen_Qwen3-32B, task_id=26, model=32B	Agent selects the wrong tool (**transfer_to_human_agents**) for the task; the reference solution begins with **find_user_id_by_email** and a different tool sequence.

3	group=retail::react::user=Qwen_Qwen3-32B, task_id=12, model=32B	Agent selects the wrong tool (**transfer_to_human_agents**) for the task; the reference solution begins with **find_user_id_by_email** and a different tool sequence.
4	group=retail::react::user=Qwen_Qwen3-32B, task_id=12, model=4B	Agent selects the wrong tool (**transfer_to_human_agents**) for the task; the reference solution begins with **find_user_id_by_email** and a different tool sequence.
5	group=airline::act::user=Qwen_Qwen3-32B, task_id=36, model=14B	Agent selects the wrong tool (**transfer_to_human_agents**) for the task; the reference solution begins with **get_reservation_details** and a different tool sequence

6.3 Examples for E3

Example #	Trajectory JSON	Why this is E{3}
1	group=airline::react::user=Qwen_Qwen3-32B, task_id=5, model=14B	Tool call sequence includes update_reservation_passengers and fails with API/tool errors.
2	group=airline::react::user=Qwen_Qwen3-32B, task_id=4, model=32B	Tool call sequence includes update_reservation_passengers and fails with API/tool errors.
3	group=retail::act::user=Qwen_Qwen3-32B, task_id=45, model=8B	Tool call get_order_details fails due to incorrect/malformed arguments.
4	group=retail::react::user=Qwen_Qwen3-32B, task_id=46, model=14B	Tool call get_order_details fails due to incorrect/malformed arguments.

5	group=retail::act::user=Qwen_Qwen3-32B, task_id=11, model=32B	Tool call find_user_id_by_email fails due to incorrect/malformed arguments.

6.4 Examples for E4

Example #	Trajectory JSON	Why this is E{4}
1	group=retail::react::user=Qwen_Qwen3-32B, task_id=47, model=8B	Agent loses state consistency: after referencing valid order IDs ['#W9502127'], it later uses placeholder #W0000000, which contradicts prior context and corrupts later decisions.
2	group=retail::react::user=Qwen_Qwen3-32B, task_id=90, model=14B	Agent loses state consistency: after referencing valid order IDs ['#W2273457', '#W8661412', '#W9284598'], it later uses placeholder #W0000000, which contradicts prior context and corrupts later decisions.
3	group=retail::act::user=Qwen_Qwen3-32B, task_id=61, model=4B	Agent loses state consistency: after referencing valid order IDs ['#W3973757', '#W5061109', '#W5797116', '#W5797164'], it later uses placeholder #W0000000, which contradicts prior context and corrupts later decisions.
4	group=retail::act::user=Qwen_Qwen3-32B, task_id=45, model=8B	Agent loses state consistency: after referencing valid order IDs ['#W9502127'], it later uses placeholder #W0000000, which contradicts prior context and corrupts later decisions.

5	group=retail::act::user=Qwen_Qwen3-32B, task_id=41, model=14B	Agent loses state consistency: after referencing valid order IDs ['#W0000100'], it later uses placeholder #W0000000, which contradicts prior context and corrupts later decisions.

6.5 Examples for E5

Example #	Trajectory JSON	Why this is E{5}
1	group=retail::react::user=Qwen_Qwen3-32B, task_id=65, model=14B	Agent violates hard domain constraints (payment/eligibility/status rules) and still attempts the constrained action path, instead of replanning payment/tool strategy.
2	group=retail::act::user=Qwen_Qwen3-32B, task_id=42, model=14B	Agent violates hard domain constraints (payment/eligibility/status rules) and still attempts the constrained action path, instead of replanning payment/tool strategy.
3	group=retail::react::user=Qwen_Qwen3-32B, task_id=92, model=32B	Agent violates hard domain constraints (payment/eligibility/status rules) and still attempts the constrained action path, instead of replanning payment/tool strategy.
4	group=airline::act::user=Qwen_Qwen3-32B, task_id=11, model=14B	Agent violates hard domain constraints (payment/eligibility/status rules) and still attempts the constrained action path, instead of replanning payment/tool strategy.

5	group=airline::react::user=Qwen_Qwen3-32B, task_id=36, model=8B	Agent violates hard domain constraints (payment/eligibility/status rules) and still attempts the constrained action path, instead of replanning payment/tool strategy.

6.6 Examples for E6

Example #	Trajectory JSON	Why this is E{6}
1	group=retail::react::user=Qwen_Qwen3-32B, task_id=65, model=14B	Agent remains in an elicitation loop (25 question marks, 62 turns) with no executable tool progress. The conversation stalls before required actions are completed.
2	group=airline::react::user=Qwen_Qwen3-32B, task_id=36, model=8B	Agent remains in an elicitation loop (28 question marks, 62 turns) with no executable tool progress. The conversation stalls before required actions are completed.
3	group=retail::act::user=Qwen_Qwen3-32B, task_id=97, model=14B	Agent remains in an elicitation loop (24 question marks, 62 turns) with no executable tool progress. The conversation stalls before required actions are completed.
4	group=retail::act::user=Qwen_Qwen3-32B, task_id=69, model=32B	Agent remains in an elicitation loop (21 question marks, 62 turns) with no executable tool progress. The conversation stalls before required actions are completed.

5	group=retail::act::user=Qwen_Qwen3-32B, task_id=113, model=4B	Agent remains in an elicitation loop (45 question marks, 62 turns) with no executable tool progress. The conversation stalls before required actions are completed.

6.7 Examples for E7

Example #	Trajectory JSON	Why this is E{7}
1	group=retail::react::user=Qwen_Qwen3-32B, task_id=9, model=8B	After receiving explicit tool errors, the agent continues with success-like language or consequential writes without proper recovery. This is an error-interpretation failure that propagates invalid state.
2	group=retail::react::user=Qwen_Qwen3-32B, task_id=38, model=14B	After receiving explicit tool errors, the agent continues with success-like language or consequential writes without proper recovery. This is an error-interpretation failure that propagates invalid state.
3	group=retail::act::user=Qwen_Qwen3-32B, task_id=42, model=14B	After receiving explicit tool errors, the agent continues with success-like language or consequential writes without proper recovery. This is an error-interpretation failure that propagates invalid state.
4	group=retail::react::user=Qwen_Qwen3-32B, task_id=103, model=32B	After receiving explicit tool errors, the agent continues with success-like language or consequential writes without proper recovery. This is an error-interpretation failure that propagates invalid state.

5	group=airline::act::user=Qwen_Qwen3-32B, task_id=11, model=14B	After receiving explicit tool errors, the agent continues with success-like language or consequential writes without proper recovery. This is an error-interpretation failure that propagates invalid state.

6.8 Examples for E8

Example #	Trajectory JSON	Why this is E{8}
1	group=retail::react::user=Qwen_Qwen3-32B, task_id=103, model=32B	Agent executes steps out of order: first tool is modify_pending_order_address instead of find_user_id_by_name_zip/find_user_id_by_email first. Prerequisite checks are skipped, increasing downstream failure probability.
2	group=airline::act::user=Qwen_Qwen3-32B, task_id=11, model=14B	Agent executes steps out of order: first tool is book_reservation instead of identity/reservation checks before write/update. Prerequisite checks are skipped, increasing downstream failure probability.
3	group=retail::react::user=Qwen_Qwen3-32B, task_id=90, model=14B	Agent executes steps out of order: first tool is cancel_pending_order instead find_user_id_by_name_zip/find_user_id_by_email first. Prerequisite checks are skipped, increasing downstream failure probability.

4	group=retail::act::user=Qwen_Qwen3-32B, task_id=90, model=8B	Agent executes steps out of order: first tool is cancel_pending_order instead find_user_id_by_name_zip/find_user_id_by_email first. Prerequisite checks are skipped, increasing downstream failure probability.
5	group=retail::act::user=Qwen_Qwen3-32B, task_id=105, model=32B	Agent executes steps out of order: first tool is return_delivered_order_items instead of find_user_id_by_name_zip/find_user_id_by_email first. Prerequisite checks are skipped, increasing downstream failure probability.

7. Individual Contribution

Student	Contribution
Nihaarika Agarwal	Pass 3
Sanyam Jain	Pass 2
Abraham Lozano Serna	Pass 1
Omkar Satish Korate	Pass 4